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Level 5, 1st Sem / DVOC (Auto, Servicing, AMT)

Subject : Modern Electric and Hybrid Vehicles

Time : 2 Hrs. M.M. : 50

SECTION-A

Note: Very short answer Questions. Attempt all ten questions. (10x1=10)

Q.1 A basic electrical circuit consists of a power source, load, and_________.

Q.2 A generator converts mechanical energy into ______ energy.

Q.3 One disadvantage of a parallel hybrid drive train is its complex ______ system.

Q.4 Plug-in hybrid vehicles can be charged from an external ______ source.

Q.5 Regenerative braking allows power to flow from the wheels back to the______ during deceleration.

Q.6 In a hybrid system, power flow is controlled based in driving conditions using embedded________.

Q.7 Brushless DC motors offer higher______ and lower maintenance than brushed motors.

Q.8 A power converter changes the voltage and current levels suitable for the_______ operation.

Q.9 Piezoelectric materials generate electrical energy when subjected to_____ stress.

Q.10 Shock absorbers in regenerative systems can be used to harvest _______ energy from road vibrations.

SECTION-B

Note: Short answer type questions. Attempt any six questions out of Eight questions. (6x5=30)

Q.11 Differentiate between electric vehicles and hybrid electric vehicles in terms of power source, efficiency, and emissions.

Q.12 Discuss the importance of electrical basics like current, voltage, and resistance in understanding EV systems.

Q.13 Explain the concept of electric traction and its application in urban electric vehicles.

Q.14 Describe the role of drive train topologies in achieving energy efficiency.

Q.15 Why regenerative braking system is required? Explain.

Q.16 Explain the working principle of DC motor.

Q.17 Explain the basic principle and working of a regenerative braking system used in electric vehicles.

Q.18 Explain in brief about the use of shock absorbers as vibration energy harvesters.

SECTION-C

Note: Long answer type questions. Attempt any one questions out of two questions. (1x10=10)

Q.19 Describe the different types of electric and hybrid electric drive train topologies with neat block diagrams.

Q.20 Describe in detail the power flow control in various hybrid drive train topologies (Series, Parallel, Series-Parallel). Use neat diagrams,